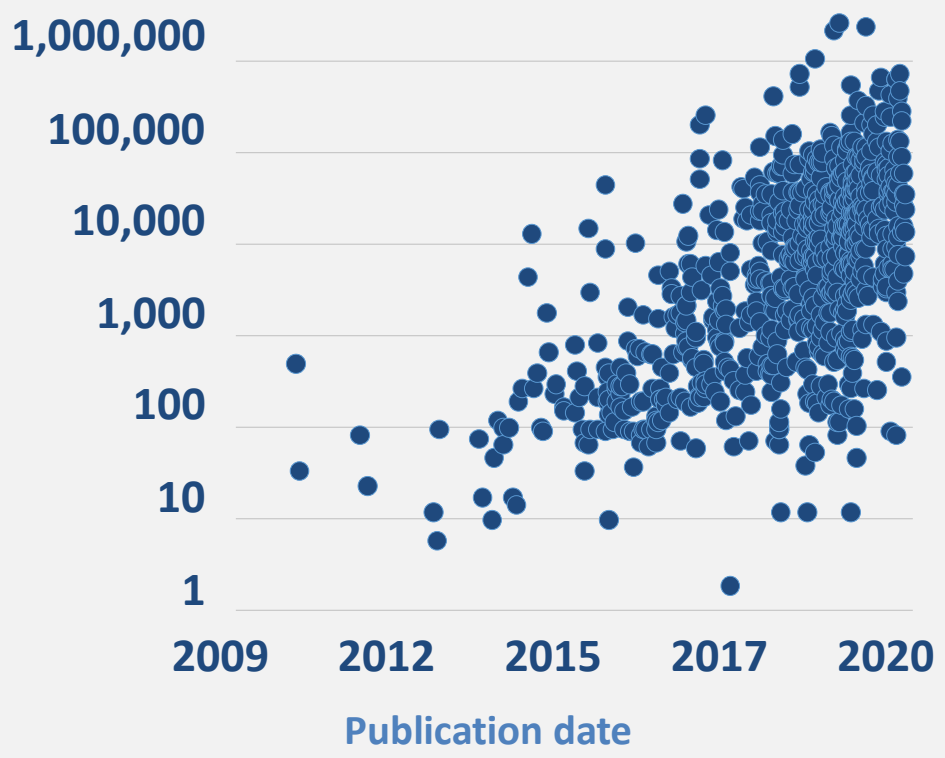
GENOMIC ANALYSIS AT THE SPEED OF MEMORY

The extra-exponential growth of data is crushing IT infrastructure used for single-cell sequencing. Big Memory Computing is designed specifically for fast processing of big data and it's already paying-off for organizations like Analytical Biosciences and TGen.

Extra-Exponential Growth of Cell Data (Watch the video)

Number of Cells per Study



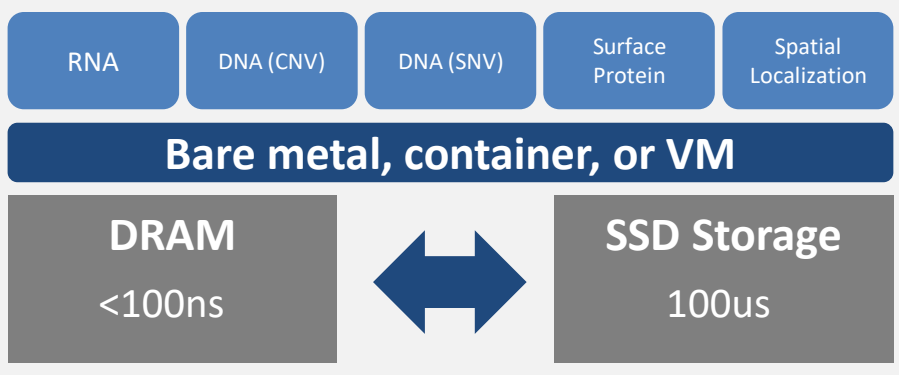
Modalities of Data

- RNA
- DNA (CNV)
- DNA (SNV)
- DNA (immune repertoire)
- Chromatin Accessibility
- Surface Protein
- Spatial Localization
- Etc.

Crushing Storage & Time-to-Insight

Because of the extra-exponential growth of cell data, terabyte data sets take a long time to load from storage into memory, and they often don't fit into memory. Storage kills time-to-insight because even SSD is 1,000 slower than memory.

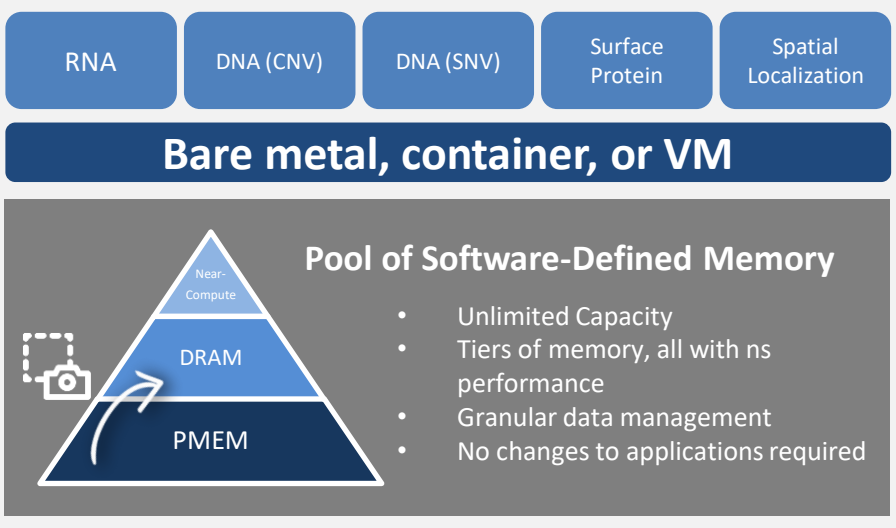
Von Neumann Computing Model



Genomic pipelines crush time-to-insight when data is greater than memory resulting in extensive IO to slower storage

Single-Cell Sequencing at the Speed of Memory

Big Memory Computing Model

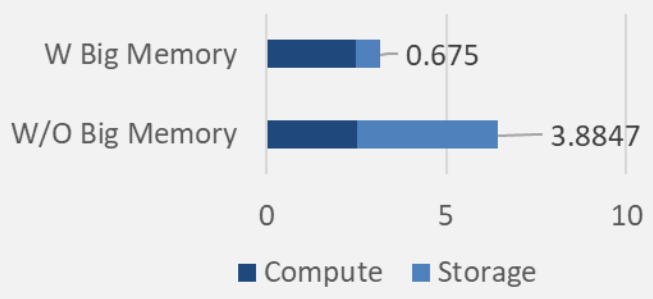


Big Memory Computing provides a massive pool of memory, so all cell data fits in memory. Gives bioinformaticians fine-grained control of in-memory snapshots for multi-threading and instant restore from a crash. Watch the video.

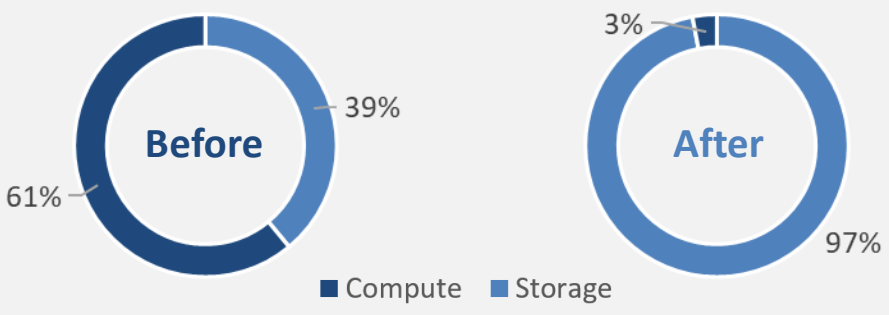
Faster Time-to-Insight (watch the video)

Analytical Biosciences

61% Faster Time-to-Insight with Big Memory



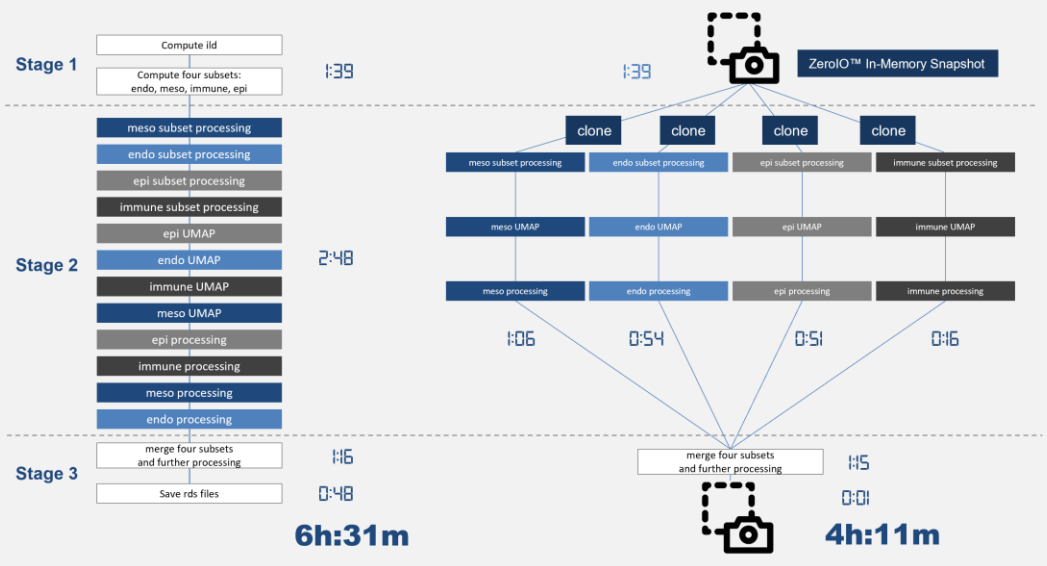
Storage IO Reduced to 3% of Time



Transforming Pipelines from Single-Threaded to Multi-Threaded

tgen AN AFFILIATE OF City of Hope

Time-to-Insight 35% Faster with Big Memory



Big Memory nodes configured so all cell data could fit in memory. In-Memory snapshots were used to create clones of the cell data set and transform the "stubbornly single-threaded" pipeline into a multi-threaded pipeline. Watch the video.

Big Memory is Intel® Optane™ Persistent Memory and Memory Machine™ software from MemVerge® working together accelerate time-to-insight. Learn more at www.memverge.com.

MemVerge